# WHICH CONDITIONING RECORDING DOES A VOICE CLONE FOLLOW? A BALANCED CROSS-MICROPHONE CROSSOVER

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## ABSTRACT

Voice-clone attribution usually stops at the generator or speaker. We ask a finer question: which of two recordings of the same speaker conditioned the clone? Under an internally frozen protocol, a fixed 54-speaker VCTK roster provides two events with paired microphone captures. Four publicly available cloning systems and four fixed, held-out texts produce 3,456 clones. Generation uses one microphone; attribution compares the two events through the other, so speaker identity, generated output text, exact waveform and a shared microphone fingerprint cannot decide the label. With fixed speaker-verification readouts, the mic1→mic2 follow rate is .896 [.869,.921] for ECAPA and .622 [.593,.652] for WavLM; the predeclared reverse direction gives .907 [.881,.931] and .620 [.589,.653]. The ECAPA result meets the predeclared large-effect criterion; a second fixed SV readout has the same aggregate direction, which persists after swapping microphones. This establishes an event-linked closed-set signal between these two fixed pickup paths, not the carrier mechanism or open-set presence detection.

***Index Terms*—** voice cloning, source attribution, audio provenance, speaker verification

## 1. INTRODUCTION

Zero-shot speech generation conditions on a short reference recording. When that recording was used without consent, identifying the generator or cloned speaker does not answer the narrower provenance question: *which recording was used?* Existing source tracing attributes a synthesis pipeline [1, 2], and clone attribution identifies an enrolled speaker [3, 4, 5]. Source-voiceprint restoration, contrastive source-speaker tracing and leakage measurement recover or quantify the source speaker after voice conversion [6, 7, 8]; they do not identify which of two recordings of that same speaker conditioned inference. Recording-level evidence would instead connect an output to one conditioning event among other recordings of the same voice.

That target is easy to confound. A clone and candidate share speaker identity; one candidate may be intrinsically closer to every clone; recording session, duration or microphone can reveal the answer; and generated content may overlap. Prior prompt-transfer work shows preservation of prosody and acoustic environment [9, 10, 11, 12], while speaker embeddings themselves encode session and channel [13, 14]. High retrieval from a same-speaker pool therefore does not by itself show that the target follows the recording used for conditioning.

We turn the question into a balanced intervention. For each speaker, clone from recording event A and from event B, and test both signed comparisons. Prompt and candidate are paired captures from different microphones. Four systems and texts are fully crossed; ECAPA-TDNN [15] is fixed as primary and a speaker-verification-fine-tuned WavLM [16] as cross-encoder corroboration. Within a 54-speaker complement inherited from ECAPA-informed development rosters, all speakers are retained and pair selection is frozen from metadata alone, without clone outcomes.

Our contributions are: (1) a complete A/B crossover that tests whether a clone follows its conditioning event while changing pickup path; (2) a complete 3,456-clone evaluation on 54 speakers, four systems and two fixed SV readouts, with public score-level and roster-selection verification; and (3) a boundary result: event attribution is consistent in a two-candidate closed set, but neither the physical carrier nor deployable recording-presence detection follows.

## 2. RELATED WORK

**Attribution and provenance.** Generator tracing, shared-generator verification and speaker attribution operate above the individual recording [1, 2, 3, 4]. Watermarking requires instrumentation before misuse [17]; membership inference asks whether audio entered training [18], not whether it was the inference-time prompt. To our knowledge, no prior controlled study intervenes on the prompt recording within speaker and then replaces the candidate pickup path.

**Prompt transfer and nuisance information.** Prompt-conditioned systems preserve prosody and environment [10, 11, 19]; prompt cosine is also a standard synthesis metric [20]. Concurrent work separates prompt environment from identity [21]. These results motivate recording-level leakage, but do not make it identifiable: the same speaker and session can support both candidates. Our design targets the *choice between A and B*, not a generic clone–prompt resemblance.

## 3. CROSS-MICROPHONE CROSSOVER

### 3.1. Task and design

For speaker $s$, let recording events $A$ and $B$ each have paired VCTK mic1/mic2 captures [22]. A clone $c_{A,m1}$ is conditioned on event A's mic1 capture. The primary comparison asks whether a fixed encoder $f$ gives

$$I[\cos(f(c_{A,m1}), f(A_{m2})) > \cos(f(c_{A,m1}), f(B_{m2}))]\,, \qquad (1)$$

with ties worth $1/2$. We also clone from B and reverse the correct label. Thus an encoder's unconditional preference for A cannot create

**Primary: opposite-pickup attribution**

$A_{m1} \to c_{A,m1} \to \{A_{m2}, B_{m2}\}$ and

$B_{m1} \to c_{B,m1} \to \{A_{m2}, B_{m2}\}$

Correct labels are reversed across arms; speaker, system and generated text are held fixed. The direction-reversal analysis swaps $m1$ and $m2$.

**Fig. 1**. Complete two-arm crossover. Generation never consumes the microphone used for the attribution candidates.

**Table 1**. What the crossover blocks, and the residual scientific boundary.

| Shortcut | Design response | Residual |
|---|---|---|
| Speaker identity | A/B within speaker | none for the label |
| Candidate preference | both signed arms | arm interactions |
| Generated content | four texts shared A/B | text fixed, not sampled |
| Exact waveform/device | opposite microphone | event acoustics |
| Duration/text length | metadata-matched A/B | other local cues |

a positive effect. The predeclared direction reversal swaps microphones (mic2 prompt, mic1 candidates) and cannot rescue a failed primary direction.

More explicitly, define $d_X = s(c_X, A) - s(c_X, B)$ after replacing candidates by the opposite-microphone captures. A system/text cell contributes $\{I[d_A > 0] + I[d_B < 0]\}/2$. A clone-independent bias toward A makes both $d_A, d_B$ positive and contributes exactly chance, rather than masquerading as seed following. Ties inherit weight $1/2$ inside each indicator.

The threat model is deliberately narrow: the analyst has a clone and two known, same-speaker candidate recordings, exactly one of which supplied the prompt. The objective is source attribution, not deciding whether an arbitrary recording was used. Encoder parameters, score direction and the tie rule are fixed before evaluation; the analyst does not train on these speakers or systems.

### 3.2. Frozen roster and generation

VCTK v0.92 documents DPA 4035 and Sennheiser MKH 800 microphones in a fixed hemi-anechoic setup [22]. Its public description does not establish acquisition timing, so our claim requires paired captures, not simultaneity. The frozen manifest independently supports the pairing: all 108 selected event/microphone pairs have identical frame counts and sampling rates.

The 54-speaker roster is the exact complement of two earlier development tiers. Tier 1 applied an ECAPA-geometry gate to 105 speakers and retained the lexicographically first 30 passing speakers; Tier 2 used ECAPA within-speaker geometry and retained all 24 remaining qualifiers. Consequently, the present roster is disjoint from development and blind to EXP-205 outcomes, but not embedding-naive or population-sampled. Within that fixed complement, all 54 speakers remain and the A/B-pair rule below never inspects an embedding, score or clone outcome.

For each speaker, selection enumerates 4–10 s paired utterances, excludes a previously used seed, and requires the A/B duration gap to be at most .25 s and the relative seconds-per-UTF-8-byte duration gap at most 5% on *both* microphones. A deterministic metadata-only rule chooses the closest pair; no embedding, score or clone outcome enters within-roster pair selection. Transcripts A and B differ, but generation uses four fixed texts disjoint from both candidates and fully crossed across arms. The selected roster is substantially tighter than the eligibility limits: its worst duration gap is .112 s and worst relative duration-per-byte gap is 2.57% across either microphone.

**Table 2**. Speaker-weighted follow rate [95% composition-stability interval]. Same-mic columns are diagnostics, never verdict inputs.

| Prompt→candidate | Encoder | Cross-mic | Same-mic diagnostic |
|---|---|---|---|
| mic1→mic2 | ECAPA | **.896 [.869,.921]** | .938 [.920,.955] |
| | WavLM | **.622 [.593,.652]** | .631 [.602,.660] |
| mic2→mic1 | ECAPA | **.907 [.881,.931]** | .930 [.910,.948] |
| | WavLM | **.620 [.589,.653]** | .659 [.627,.692] |

We evaluate F5-TTS 1.1.22 [23], XTTS-v2 0.25.3 [24], CosyVoice 2 [25] and Seed-VC [26]. Seed-VC converts one fixed held-out LibriTTS-R source utterance per generation text, shared across targets [27]. The full design is 54 speakers $\times 2$ events $\times 2$ prompt microphones $\times 4$ texts $\times 4$ systems $= 3{,}456$ clones. Exact checkpoints, repositories, environments, files and per-clone ancestry ledgers are hash-pinned.

### 3.3. Estimand, bars and provenance

Each speaker contributes 32 dependent comparisons per direction (two arms, four systems, four texts). We first average within speaker, then give all 54 speakers equal weight. Percentile intervals resample whole speaker vectors 100,000 times with a frozen seed. They quantify stability to speaker composition in this fixed eligible roster, not population-coverage confidence. Systems and texts are fixed crossed factors; system-specific points are descriptive and receive no individual inferential claims.

The frozen primary conjunction is: **E**, ECAPA lower endpoint $> .50$; **O**, ECAPA point $\geq .80$ and lower endpoint $> .70$; and **R**, WavLM lower endpoint $> .50$. “O” is the protocol’s internal label; it is an author-declared large-effect criterion, not a deployment-utility claim. The direction-reversal headline is permitted only if both reverse ECAPA and WavLM lower endpoints exceed .50. Diagnostics cannot rescue any conjunct.

Contemporaneous project records state that the manifest, analyzer, verdict truth table and execution config were frozen before outcomes, but they have no external timestamp and are not presented as public preregistration. The analyzer authenticates the complete clone/candidate census before parsing similarities. A transport-only manifest-object mismatch stopped fail-closed; its one-field repair is documented in the same internal history. A separate checker rehashes 3,456 clone receipts plus 216 candidates and reconstructs every result field to $2 \times 10^{-15}$. The portable artifact contains sanitized scores, logical manifests, original trust-root hashes, documented source adaptations and a verifier for the 3,456-row grid, 100,000 bootstraps and verdict. This is score-level reproducibility, not proof of audio-to-score ancestry without the third-party audio and model bytes.

## 4. RESULTS

### 4.1. The conditioning event survives pickup change

Table 2 reports the predeclared directions. ECAPA attributes the conditioning event in 89.6% of primary comparisons and 90.7% in reverse. The primary point and lower endpoint meet the .80/.70 predeclared large-effect criterion; the reverse ECAPA interval is wholly above chance. WavLM is much weaker but directionally consistent: 62.2% and 62.0%, with both intervals wholly above chance. Therefore E, O, R and both reverse components pass. Under the frozen rule, the ECAPA effect is large, WavLM provides a second-fixed-SV-readout corroboration, and the direction persists after swapping microphones.

**Table 3**. Frozen bars and observed values. LCB is the lower endpoint in Table 2; inequalities are strict except the .80 point bar.

| Axis | Frozen criterion | Observed | Status |
|---|---|---|---|
| E | primary ECAPA LCB $>$ .50 | .869 | PASS |
| O | ECAPA point $\geq$ .80, LCB $>$ .70 | .896/.869 | PASS |
| R | primary WavLM LCB $>$ .50 | .593 | PASS |
| $D_E$ | reverse ECAPA LCB $>$ .50 | .881 | PASS |
| $D_R$ | reverse WavLM LCB $>$ .50 | .589 | PASS |

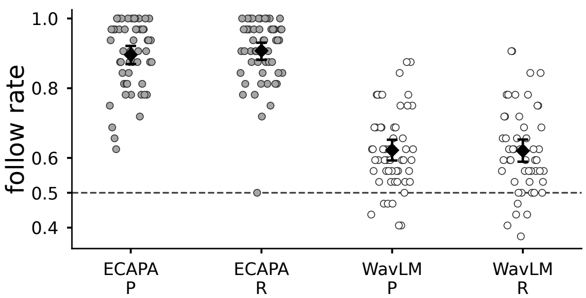


**Fig. 2**. Fixed-roster speaker means (54 dots/group; 32 comparisons per dot). Diamonds and bars are pooled points and 95% composition-stability intervals; the dashed line is chance. P/R denote primary/reverse directions.

The same-microphone diagnostics are only modestly higher, so the signal is not created by an identical pickup path. This does not prove microphone invariance: both microphones observe the same event, and device-specific transformations can interact differently with the two encoders. It does show that exact microphone identity shared between prompt and candidate is unnecessary.

Figure 2 exposes the heterogeneity hidden by pooled points. All 54 primary ECAPA speaker means exceed .50; in reverse, 53 exceed it and one equals it. WavLM is not uniformly positive—47 primary and 43 reverse speaker means exceed .50—so its contribution is corroboration of the aggregate direction, not a per-speaker guarantee. Whole-speaker resampling preserves this heterogeneity rather than treating 1,728 comparisons per direction as independent.

### 4.2. Breadth across fixed systems

Table 4 gives descriptive system points. ECAPA exceeds .82 for every system and direction; WavLM ranges from .572 to .671. The pooled claim is not obtained by selecting the best generator, and the weaker XTTS-v2 cells remain visible. Because systems were fixed rather than sampled and no simultaneous per-system intervals were planned, we claim breadth of the tested design, not a universal system effect.

The ECAPA–WavLM gap is scientifically useful. It rules out a claim that every speaker representation exposes the event-linked signal at near-ceiling magnitude. WavLM's lower endpoints show the same aggregate direction under a second fixed SV readout, but do not establish representation independence or exclude sensitivities shared by SV encoders. Continuous margins are positive in the same directions, but were diagnostic and do not change the binary result.

### 4.3. Post-result sensitivity checks

An ancestry audit found that the evaluation complement inherits ECAPA-informed development selection. Table 5 reports a post-hoc descriptive stratification by the earlier Tier 1 gate. The direction persists among both the 36 pass-but-not-retained and 15 fail speakers. Three roster speakers were absent from that gate record and are kept separate, not assigned a status. This check quantifies the inherited composition threat; it cannot reconstruct an embedding-naive sample or alter the frozen pooled verdict.

**Table 4**. Descriptive follow-rate points by system. P/R: primary/reverse.

| Encoder, dir. | F5-TTS | XTTS-v2 | CosyVoice 2 | Seed-VC |
|---|---|---|---|---|
| ECAPA, P | .942 | .833 | .910 | .898 |
| ECAPA, R | .942 | .822 | .942 | .924 |
| WavLM, P | .618 | .572 | .671 | .627 |
| WavLM, R | .657 | .574 | .657 | .593 |

**Table 5**. Post-hoc points by prior Tier 1 gate status. P/R: microphone direction; no stratum-specific confirmatory claim.

| Prior gate status | $n$ | ECAPA P/R | WavLM P/R |
|---|---|---|---|
| Pass, not retained | 36 | .908/.907 | .609/.596 |
| Fail | 15 | .879/.915 | .654/.685 |
| Not evaluated | 3 | .833/.875 | .625/.583 |

Arm-separated points also oppose a one-candidate shortcut. For A/B, ECAPA is .889/.903 in primary and .892/.922 in reverse; WavLM is .588/.656 and .660/.581, respectively. Thus WavLM's arm asymmetry reverses with microphone direction. All 32 system$\times$direction$\times$encoder$\times$arm cells are above .50 (minimum .523), descriptively; no cell has a separate interval or multiplicity-adjusted test.

### 4.4. What changed relative to earlier evidence

Earlier same-speaker mixed-chapter pools retrieved the seed at 9.7–16.4$\times$ chance, and same-chapter fused AUC was .815–.872. Those designs established a floor but left seed and speaker/session structure entangled; F5-TTS also exposed a duration-per-text-byte cue. A subsequent two-seed crossover obtained .806 [.757,.854] but missed its predeclared .90 point bar and used favorable seed pairs on one pickup path. We retain that as positive discovery, not confirmation.

The present experiment is a separate test: the 54-speaker development complement, all speakers retained, within-roster metadata-only A/B pairs, balanced labels, opposite-microphone candidates and fixed readouts. Its .80/.70 ECAPA bar was frozen for this design; passing it does not retroactively change the earlier .90 miss. This trajectory matters: each control removed a shortcut rather than merely adding another positive cell.

## 5. BOUNDARY AND LIMITATIONS

**Attribution is not presence detection.** The crossover is a closed-set two-alternative question known to contain the source. In our earlier paired present/absent evaluation, normalized minDCF at a .01 prior was .79–1.00 against 1.00 for reject-all; only two of four systems had intervals excluding reject-all. A score can order A above B while lacking a stable global threshold. We therefore propose investigative triage, not an automated accusation or a claim that a queried recording is absent.

**The carrier is unidentified.** Opposite microphones exclude an exact waveform and shared device fingerprint, while A/B balancing excludes a fixed candidate preference. Metadata matching sharply limits raw duration and text-normalized-duration shortcuts. It does not separate candidate transcript, event-level prosody, articulation, background acoustics, paired-capture room state or their interactions with model conditioning. F5-TTS and CosyVoice use the reference transcript whereas XTTS-v2 and Seed-VC do not; positive points in all four systems show transcript is not a complete explanation, but it remains a system-dependent carrier. A mechanism claim requires interventions on these factors.

**Scope.** The result concerns four fixed checkpoints, four fixed texts, read English speech and one inherited 54-speaker VCTK complement. It is disjoint from the earlier development rosters but ECAPA-informed through their composition, not globally unseen or population-sampled. Checkpoint training corpora are not documented at recording level, so VCTK exposure cannot be excluded. The intervals do not sample systems, texts or a speaker population. Natural speech, noisy field recordings, other pickup paths, more than two candidates and open-set search remain untested.

**Release and misuse.** We release code, hashes, provenance metadata, scores and aggregate/per-speaker results, but not model weights, source audio or playable impersonations. Recording attribution can assist provenance audits and also deanonymize speakers; licensing alone does not settle consent.

**Compliance with Ethical Standards.** This study analyzes licensed public speech and synthetic outputs; no participants were recruited. We release no source or generated audio and claim no institutional ethics determination.

## 6. CONCLUSION

In this fixed roster, a clone carries a signal linked to which same-speaker event conditioned it when prompt and candidates use the two tested microphone paths. The effect meets the predeclared large-effect criterion with ECAPA, has the same weaker aggregate direction under WavLM, and persists after reversing microphones while pooling equally over four fixed systems. This is a large closed-set signal between two tested paths; it neither identifies the carrier nor solves open-set recording-presence detection.

## 7. REFERENCES


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